T.C.
YILDIZ TEKNİK ÜNİVERSİTESİ
ENDÜSTRİ MÜHENDİSLİĞİ BÖLÜMÜ

Innovation Campus Artificial Intelligence (Samsung)
Group 13

Mehmet Can Demirkol 21061041
Anıl Danacı 22069922
Selen Bilici 24061620
Arzum Doğan 22061048

Project Name

AI-Based Exploratory Data Analysis of Global Hunger Index

Overview

SDG 2: Zero Hunger is one of the most important worldwide issues that this project seeks to address. We use the Global Hunger Index (GHI) 2025 dataset to conduct a thorough Exploratory Data Analysis (EDA) to identify trends in food insecurity throughout various time periods and geographical areas. Finding the countries most at risk and comprehending the underlying indicators, such as malnutrition and infant mortality, that influence these hunger scores are the main goals.

We convert unprocessed institutional data into useful insights by using Python data cleaning and visualization tools. Future AI models that can anticipate famine crises before they worsen will be built on this analysis. Our study promotes data-driven decision-making for more effective resource allocation in sustainable development initiatives by drawing attention to these trends.

Background

One of the biggest obstacles to global sustainable development is still food insecurity. Millions of people continue to experience chronic hunger despite worldwide efforts. This problem is made worse by geopolitical conflicts, economic instability, and climate change. In the past, slow, manual reporting systems were used to track hunger levels, which frequently caused humanitarian interventions to be delayed.

Our approach to this challenge has changed in recent years due to the availability of large-scale datasets like the Global Hunger Index. With indicators including undernourishment, child stunting, wasting, and death, the GHI offers a multifaceted assessment of hunger.

However, raw data alone is not enough. We must use AI and exploratory data analysis (EDA) methods to find hidden patterns and spatiotemporal trends that are not immediately apparent if we are to have a significant influence on SDG 2: Zero Hunger. We may transition from reactive to proactive, data-driven initiatives by using computational tools to analyze these trends. This background provides context for our research, in which we identify the most vulnerable populations by cleaning, processing, and visualizing the GHI 2025 data using Python-based tools.

Key Objectives / Business Objectives

Research Questions

This study aims to address the following research questions:

- Which countries have the highest and lowest hunger scores?
- How does hunger vary across different regions of the world?
- How have hunger levels changed over time?
- What patterns can be observed in the Global Hunger Index dataset?

Key Steps

The key steps undertaken in this study were as follows:

- Select the Global Hunger Index dataset.
- Clean and preprocess the dataset.
- Perform exploratory data analysis.
- Visualize key trends using graphs.
- Identify important insights related to global hunger patterns.

Methods and Workflow

Dataset

The project uses the Global Hunger Index dataset, which contains hunger scores and related indicators for multiple countries. The dataset includes different sheets such as country rankings, indicator values, and overall hunger scores.

Data Cleaning and Preparation

The dataset was first imported into the Python environment using the Pandas library. Pandas provides efficient tools for structured dataset loading, organization, and exploration.

The first few rows of the dataset were displayed using the *df.head()* function once the dataset had been loaded. This phase made it possible for us to rapidly examine the data's structure, including the names of the columns and the kinds of information they included.

```python
from google.colab import drive
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

drive.mount('/content/drive')
file_path = '/content/drive/MyDrive/Samsung/ÖdevGrup13/2025.xlsx'

df_drive = pd.read_excel(file_path)
xls = pd.ExcelFile(file_path)
print(xls.sheet_names)
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
['GHI Ranking 2025', 'GHI Indicator Values 2025', 'GHI Scores 2025 ']
```

Next, the overall structure of the dataset was examined using the *df.info()* method. This function provides important information about the dataset, including the number of entries, the number of columns, the data types of each column, and the presence of missing values. By using this method, we were able to gain an initial understanding of the dataset's structure and quality before proceeding with further analysis. The results indicated that the dataset contains approximately 140 entries and 7 columns, representing Global Hunger Index scores and related information for different countries. This step also helped identify potential issues, such as missing values, that might need to be addressed during the data cleaning process.

The *df.isnull().sum()* function was then used to check for missing values. The findings indicated that while there are about 3 missing entries in a few columns, the dataset is still mostly full and appropriate for analysis.

Lastly, simple descriptive statistics were produced using the *df.describe()* function. This stage gave an overview of the distribution of values across the columns and assisted in summarizing the dataset. The summary primarily showed data like the number of unique values and the most frequent entries because many columns were kept as object types.

```python
# veri bilgisi
df.info()

# eksik değerler
df.isnull().sum()

# temel istatistikler
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 140 entries, 0 to 139
Data columns (total 7 columns):
 #   Column                                                      Non-Null Count  Dtype
---  ------                                                      --------------  -----
 0   2000, 2008, 2016, AND 2025 GLOBAL HUNGER INDEX SCORES, AND CHANGE SINCE 2016  139 non-null   object
 1   Unnamed: 1                                                  137 non-null    object
 2   Unnamed: 2                                                  137 non-null    object
 3   Unnamed: 3                                                  137 non-null    object
 4   Unnamed: 4                                                  137 non-null    object
 5   Unnamed: 5                                                  137 non-null    object
 6   Unnamed: 6                                                  137 non-null    object
dtypes: object(7)
memory usage: 7.8+ KB
```

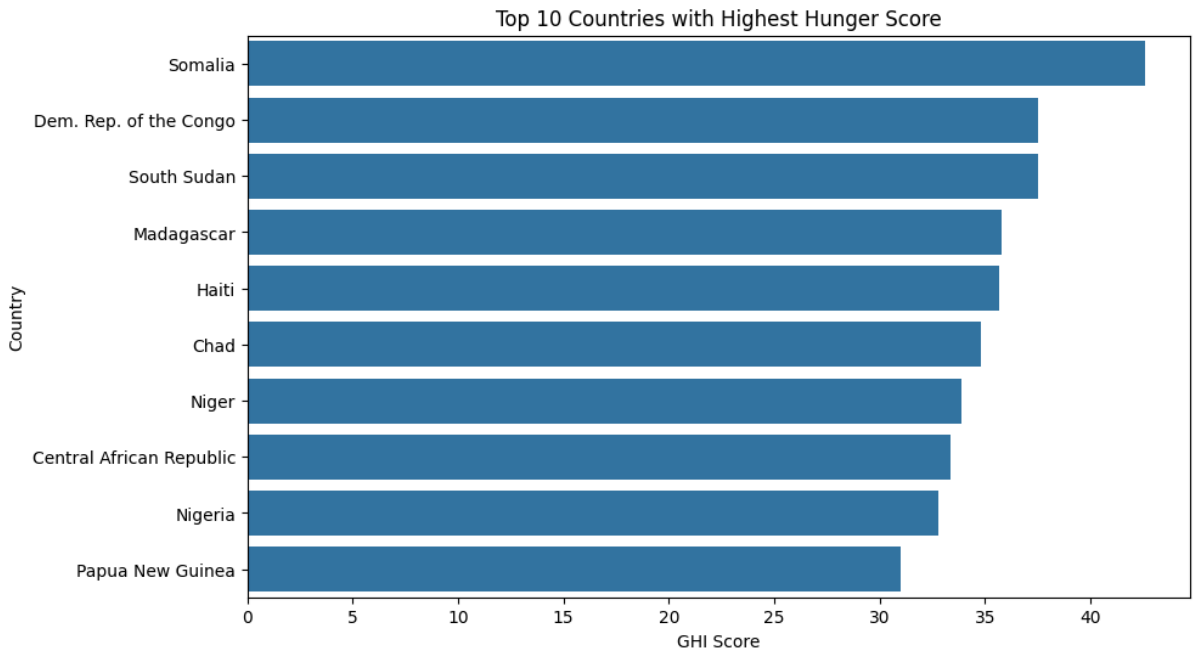
	2000, 2008, 2016, AND 2025 GLOBAL HUNGER INDEX SCORES, AND CHANGE SINCE 2016	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6
count	139	137	137	137	137	137	137
unique	139	102	100	88	87	82	94
top	Country\n with data from	—	<5	<5	<5	—	—
freq	1	15	12	18	25	39	39

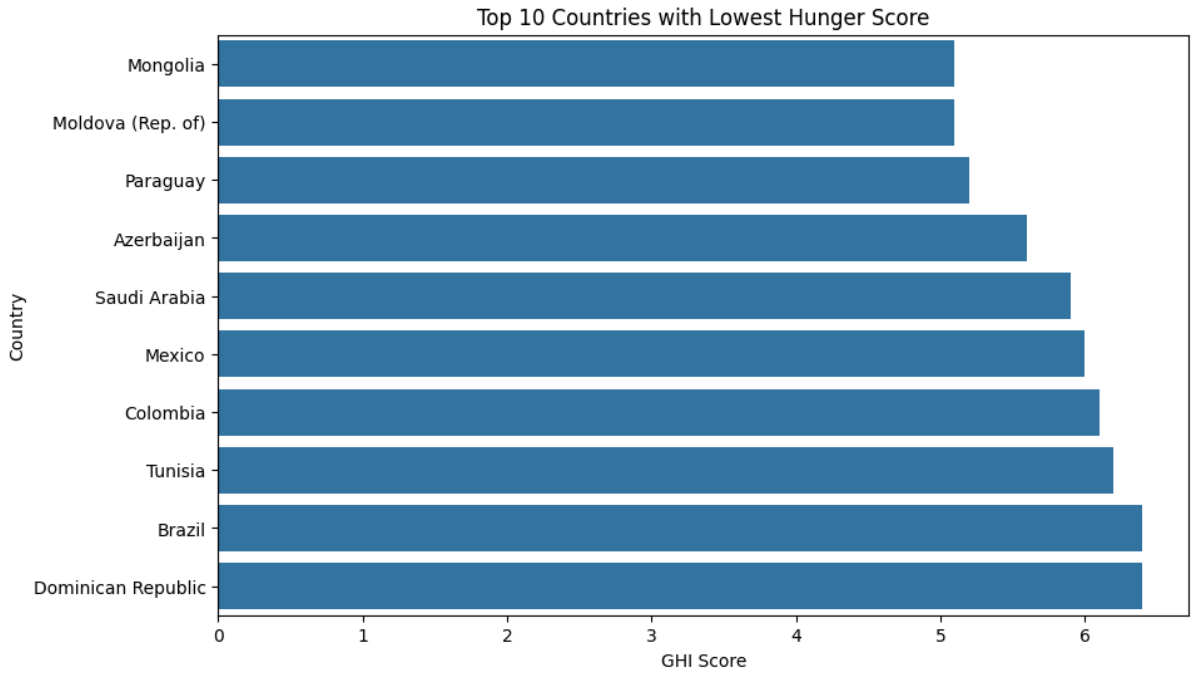
Modelling

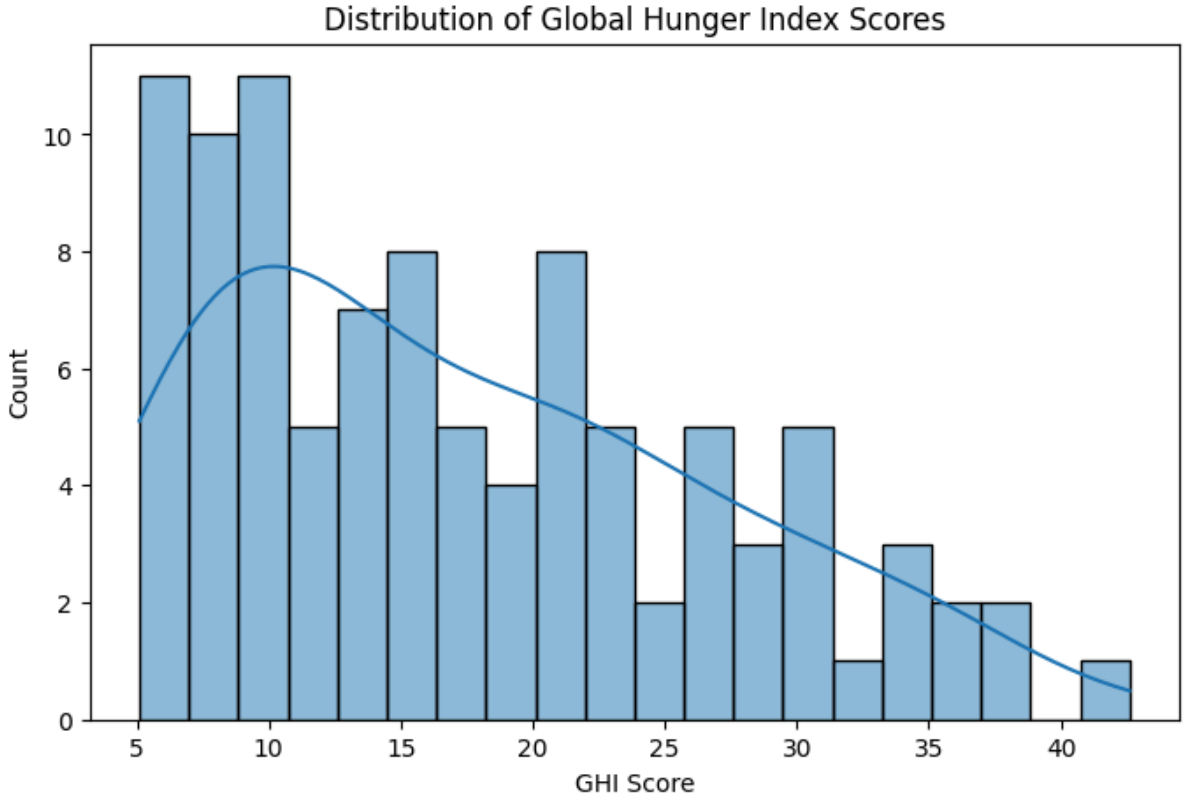
- We will calculate moving averages and identify growth rates in hunger scores over the past decades.
- A linear Regression approach will be applied to the plot current data trends and project estimated GHI scores for the year 2030.
- We will analyze the relationship between the four GHI indicators to determine which factor (Undernourishment, Child Stunting, Child Wasting, Child Mortality) most heavily influences the total score.
- All hunger indicators will be scaled to allow for an accurate comparative analysis between different geographic regions.

Deliverables

- A series of line graphs and bar charts showing the historical progression and predicted future of global hunger.
- Data tables ranking countries based on their projected GHI scores, highlighting those furthest from the “Zero Hunger” goal.
- A summary of key findings discovered during the EDA process, focusing on spatiotemporal patterns in food insecurity.


Top 10 Countries with Highest Hunger Score
Somalia
Dem. Rep. of the Congo
South Sudan
Madagascar
Haiti
Chad
Niger
Central African Republic
Nigeria
Papua New Guinea
Country
0
5
10
15
20
25
30
35
40
GHI Score


Top 10 Countries with Lowest Hunger Score
Mongolia
Moldova (Rep. of)
Paraguay
Azerbaijan
Saudi Arabia
Mexico
Colombia
Tunisia
Brazil
Dominican Republic
Country
0
1
2
3
4
5
6
GHI Score


Distribution of Global Hunger Index Scores
Count
GHI Score
0
2
4
6
8
10
5
10
15
20
25
30
35
40